What Should a Coding Agent Keep?

Comparing Context Compaction Artifacts for Task Continuity

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Paper & Code

Key Result: A five-field state sheet matches the full conversation history at **1.000** continuity while using **26.6% fewer tokens** and **0.52s lower latency** per call.

Abstract

AI coding agents that run for many turns eventually fill their context window. The system must **compress** older turns into a shorter memory note.

We test **five compression formats** on 30 hand-verified coding tasks and measure how well each format lets the model pick up where it left off.

A compact **five-field state sheet** (file, goal, constraint, detail, next step) achieves **perfect continuation accuracy** while cutting prompt size by 26.6%.

Freeform bullet summaries fail mainly on **file routing**: the model knows *what* to do but edits the *wrong file*.

1. Why This Matters

When a coding session is long, old messages get dropped or summarized. If the summary forgets which file was being edited—or preserves an outdated instruction—the agent's next action can go completely off track.

Research question: Given a fixed word budget, which *text format* best preserves the information an agent needs to resume correctly?

Five formats compared:

- **C0 — Full transcript**: keep everything (upper bound)
- **C1 — Recent messages only**: drop old turns (baseline)
- **C2 — Bullet summary**: freeform notes about the session
- **C3 — State sheet**: five named fields (our proposal)
- **C4 — Hybrid sheet**: state sheet + raw detail block

2. When Old Instructions Come Back to Bite

Some tasks plant a trap: the *most recent* messages repeat an outdated plan, while the actual correction sits further back in history. We call these **hard variants**.

Earlier turn: switch to `webhook_consumer.py`

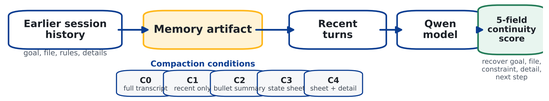
Recent turns repeat stale plan: stay in `webhook_schema.py`

Recent-only (C1)	**State sheet (C3)**
follows newest message → **FAIL**	keeps corrected file → **PASS**

On these hard tasks, truncation accuracy **drops to 0.500** and bullet summaries fall to 0.800. The state sheet stays at **1.000**—its explicit fields override the misleading recent messages.

3. How We Tested This

30	3	5	2
tasks	families	fields	models



We created 30 coding-session scenarios requiring the model to recall specific details from earlier in the conversation. Each has an **easy** version and a **hard** version (where recent turns contain outdated instructions).

Three task families (10 each):

- **Constraint updating:** an earlier plan was corrected mid-session; the model must follow the correction, not the original.
- **Exact-match retention:** success requires remembering a specific command, regex, variable name, or API key verbatim.
- **Workflow resumption:** earlier setup steps are done; the model must continue from where it stopped, not repeat finished work.

Scoring: The model picks among labeled candidates for five fields (`goal`, `file`, `constraint`, `detail`, `next_step`) via exact match. All compact artifacts share the same **85-word budget**, frozen before evaluation.

Models: `qwen3.5:4b` (primary) and `qwen3.5:9b` (robustness), via Ollama, temperature 0.

4. What a State Sheet Looks Like

A concrete example from our dataset — the state sheet stores five explicit fields vs. a bullet summary's freeform narrative:

C3 State Sheet (structured)

```
file: webhook_consumer.py
goal: add retry logic to webhook handler
constraint: max 3 retries, exponential backoff
detail: switched from schema.py after task #4
next_step: implement try/except in process_event()
```

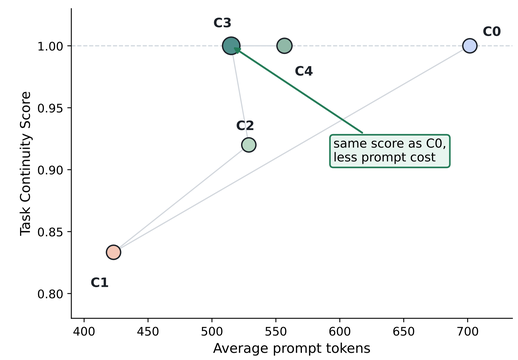
C2 Bullet Summary (freeform)

- We discussed adding retry logic to the webhook handler with exponential backoff.
- The plan is to wrap the processing function in a try/except block.
- Earlier we looked at the schema file but moved on.

× **Missing:** no filename → model edits wrong file

The state sheet's **explicit `file:` field** prevents the most common error. Bullet summaries mention "the schema file" but never record which file the agent should *actually edit next*.

5. Main Result: Same Accuracy, Fewer Tokens



−26.6%	−0.52s
prompt tokens vs. full transcript	avg. latency vs. full transcript

The state sheet (C3) matches the full conversation history at **1.000 accuracy** while using 26.6% fewer prompt tokens and finishing each call 0.52 s faster.

The hybrid sheet (C4) also scores perfectly but uses *more* tokens than C3. Adding raw detail on top of the structured fields provides no measurable benefit—better *organization* matters more than more *information*.

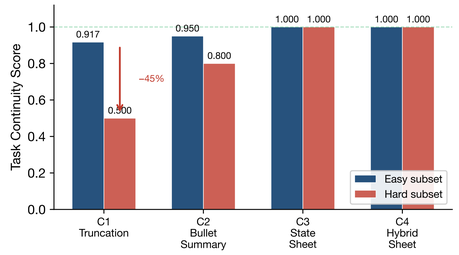
6. Where Bullet Summaries Go Wrong

Bullet summaries (C2) get the goal and next step right—but pick the **wrong file** in all 9 of their errors. The freeform notes capture the "story" but drop the specific filename.

Truncation (C1) fails more broadly: mixed state (4), lost constraints (4), wrong files (3), and lost details (2).

State sheets (C3/C4) store each fact in a dedicated field. Result: **zero errors**.

7. Robustness: Bigger Model, Same Ranking



Model	C1	C2	C3	C4
`qwen3.5:4b`	0.833	0.920	**1.000**	**1.000**
`qwen3.5:9b`	0.920	0.960	**1.000**	**1.000**

Scaling from 4B to 9B helps the weaker formats (C1 and C2 improve) but does not change the ranking. The state sheet wins at both sizes.

The hard/easy chart shows the real story: on easy tasks, even truncation scores 0.917. On hard tasks—where recent messages are misleading—it crashes to **0.500**. The memory format only matters when the context is adversarial.

8. Takeaways

Dropping old messages loses critical details, especially when recent turns are misleading.

Freeform summaries capture the overall story but forget which file to edit—a single missing anchor derails the whole action.

A five-field state sheet fixes this by giving each key fact its own slot. It matches the full transcript at 26.6% fewer tokens.

Adding more raw detail (C4) does not help—**structure beats volume**.

Bottom line: Start with a short structured schema, not a longer freeform summary.

Limitations & Future Work

30 tasks: clear trends, but too few for significance tests.

Closed-choice scoring: inflates C3/C4 by simplifying the task.

Single model family (`qwen3.5`): other architectures untested.

Single word budget (85 words): scaling behavior unexplored.

Related Work

Long-term memory: LoCoMo [1] benchmarks multi-session recall; Jia et al. [2] show recall degrades unevenly across information types.

Prompt compression: Chuang et al. [3] learn natural-language compression; EcoRAG [4] compresses retrieval context for RAG.

Agent memory: HiAgent [5] and Memory-Augmented Coding Agents [6] build hierarchical memory layers. Our work is deliberately simpler: we compare **frozen text artifacts**, not learned systems.

References

[1] Maharana et al. (2024). *LoCoMo.* ACL. [2] Jia et al. (2025). *Evaluating Memory Quality.* NAACL. [3] Chuang et al. (2024). *Learning Prompt Compression.* EMNLP. [4] Jeong et al. (2025). *EcoRAG.* ICLR. [5] Hu et al. (2025). *HiAgent.* NAACL. [6] Kang et al. (2025). *Memory-Augmented Coding Agents.* ICLR.

Code & Data: `https://github.com/NumerousJLs/context-compaction-paper`